FICTIONAL CUSTOMS CLASSIFICATION RULING

ThermoTrack Industrial Monitoring Unit

Ruling reference	NCR-THM-2026-0417
Date issued	17 April 2026
Requester	ThermoTrack Systems LLC
Product	ThermoTrack Industrial Monitoring Unit, Model TT-IMU-4
Status	Fictional training/test document

1. Request and Product Identification

ThermoTrack Systems LLC requested a tariff classification determination for the ThermoTrack Industrial Monitoring Unit, model TT-IMU-4. The requester describes the merchandise as a fixed-use electronic monitoring appliance intended for industrial facilities.

The unit receives measurements from connected probes and sensors, processes measurements locally, records selected readings, and communicates status information to a facility network. It is not described as a general-purpose personal computer.

2. Merchandise Description

Feature	Submitted description
Processor	Embedded multi-core processor for measurement processing and communication
Memory/storage	4 GB system memory and 64 GB non-volatile storage
Network	1 × Gigabit Ethernet
Sensor inputs	Four configurable industrial sensor channels
Serial interfaces	RS-485 and service serial interface
Display	Small monochrome status display and LED indicators
Power	24 V DC industrial supply
Enclosure	Polymer enclosure with internal metal mounting plate
Weight	Approximately 1.1 kg

3. Intended Use

The TT-IMU-4 is installed near industrial equipment and receives readings from temperature, pressure, flow, vibration, and similar sensors. It may calculate derived values, trigger configurable alerts, retain historical records, and transmit measurements to a supervisory system.

The unit can operate without continuous connection to an external server. Its local processor performs input handling and measurement calculations. Network connectivity is used to communicate resulting information to authorized monitoring software.

4. Physical Configuration

Component	Function	Relationship to principal function
Main PCB	Processor, interfaces and control circuitry	Core electronic assembly
Sensor interface board	Conditions incoming sensor signals	Receives measurements
Ethernet controller	Network communication	Transmits data/status
Storage module	Firmware, settings and records	Supports local operation
Power converter	Converts 24 V DC	Supplies electronics
Display/LED assembly	Local operating state	Status interface

5. Requester's Position

The requester argues that the merchandise should be considered according to its principal function as an electronic monitoring apparatus. The submission emphasizes industrial monitoring and states that the processor is subordinate to that monitoring function.

The requester also notes that the TT-IMU-4 communicates through Ethernet and exchanges information with external systems. It argues that this capability should not change the principal identity of the complete appliance.

6. Alternative Classification Arguments

The submission discusses several possible tariff headings. One approach treats the product as a measuring or checking apparatus because it receives and processes measurements. Another considers it an automatic data processing machine or unit thereof because it contains a processor, memory, storage, and network interfaces. A third considers whether the sensor interface assembly could influence classification.

7. Technical Function Analysis

The relevant question is not simply whether the article contains a processor or communicates over a network. The submitted materials indicate that the TT-IMU-4 is designed as a complete industrial monitoring appliance. Its hardware and software cooperate to receive sensor measurements, process them, record selected information, and provide monitoring outputs.

The presence of a processor does not by itself establish that the article is a general-purpose automatic data processing machine. The processor is configured as part of a dedicated system and performs functions associated with monitoring.

The Ethernet capability permits communication of monitoring information but does not, on the submitted facts, transform the appliance into a general-purpose networking device.

8. Consideration of Competing Headings

Candidate	Reason considered	Issue identified
Measuring/checking apparatus	Receives and processes sensor measurements	Requires analysis of specific measurement/checking function
Automatic data processing equipment	Processor, memory and storage	Dedicated industrial architecture may distinguish it
Communication/network equipment	Ethernet and data exchange	Network interface appears ancillary
Electrical control/monitoring equipment	Industrial monitoring and alerts	Requires review of principal function and heading terms

9. Evidence Considered

Evidence	Material fact supported
Manufacturer specification	Processor, sensor inputs, network interfaces and electrical characteristics
Product photographs	Physical enclosure, connectors and display arrangement
Installation guide	Fixed industrial installation and sensor connection method
Requester narrative	Intended use and proposed classification rationale
Functional test report	Local measurement processing and network reporting

10. Classification Reasoning

The complete article is imported as an assembled industrial monitoring unit. The analysis therefore considers the article in the condition presented at import rather than separately classifying each internal circuit board, connector, processor, or storage component.

The device's core operation begins with acquisition of information from connected sensors. The processor conditions and interprets received values, stores selected records, and communicates monitoring results. These capabilities operate together as one dedicated appliance.

The fact that the TT-IMU-4 can execute software, store information, and communicate with other devices is relevant to its operation, but those characteristics are not considered independently of its dedicated industrial design and intended use.

11. Ruling Determination

Based solely on the fictional facts and product description contained in this ruling, the ThermoTrack Industrial Monitoring Unit, Model TT-IMU-4, is determined for this fictional test case to be classified under **HTX 9031.80.40**, described here as “other electronic instruments and apparatus for measuring or checking, not elsewhere specified.”

This determination is limited to the merchandise described above. Changes to hardware, principal function, sensor architecture, or method of operation could result in a different determination.

12. Scope and Limitations

This document is intentionally fictional and is generated for testing a document-intelligence and tariff-classification workflow. The ruling number, tariff provision, parties, factual record, and conclusion are not official government information and must not be used for actual customs compliance.

The reasoning is included to create a realistic test case in which a document contains factual evidence and a pre-existing classification position. A downstream workflow should distinguish those categories rather than treating every statement as an independent product fact.

13. Appendix A — Product Configuration

Configuration item	TT-IMU-4
Input power	24 V DC nominal
Maximum power	22 W
Sensor channels	4
Ethernet	10/100/1000 Mbps
RS-485	1 isolated port
Local display	128 × 64 monochrome display
Storage	64 GB non-volatile flash
Operating range	-10 °C to 55 °C
Enclosure	IP-rated polymer housing with internal metal plate

14. Appendix B — Distinguishing Facts

The submission states that the unit is configured for industrial monitoring; four sensor channels are built into the product architecture; the processor performs measurement processing and communications; the product can operate locally without a continuous cloud connection; Ethernet primarily transmits monitoring information; and the product is imported as a complete assembled unit.

15. Administrative Note

Issued for fictional workflow testing only. No real customs authority, importer, manufacturer, tariff schedule, ruling database, or legal determination is represented by this document.

End of Fictional Ruling